Programmable RDS Radio Receiver on ATMEGA88 Microcontroller on the Basis of RDA5807M Chip as the Central Module in Internet of Things Networks

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Abstract. The article presents innovative approach to RDS (Radio Data System). In the proposed solution, the microcontroller communicates with the system of the radio receiver via I2C bus, checking individual memory addresses, where the current data from the radio data system are stored. Selection of a radio station or adjusting of radio frequency is made through the user menu on OLED I2C display (I2C – two-way serial bus used for data transfer). The transmission form uses only two data lines: SDA (serial data) and SCL (serial clock). These two lines are used to transfer all information between devices connected to I2C bus). By means of a microswitch or infrared remote control a user may select a radio station, set an alarm and change a mobile radio set into a comfortable alarm clock. The device has a connection for additional external Wi-Fi and Bluetooth modules, and thanks to them it may become an universal, mobile central station in IOT networks. Thanks to this, a user may always have at hand a small radio set, which may be used additionally - for instance in a smart home – as the controller that switches the light on and off, check temperature in individual rooms of a factory or is used as a remote controller of a gate to the premises.

Keywords: Internet of Things · Microcontroller AVR · Radio Data System I2C transmission · C programming language

1 Introduction

The purpose of this article is to present a prototypical device, which performs functions of an RDS radio receiver and has an additional functionality – user-modifiable mobile central base in IOT networks.

At the moment there are numerous devices on the market that facilitate user's access to information, news, music, etc. and a possibility of communication among such electronic subassemblies is a kind of standard embedded in almost all basis items of household equipment, such as TV sets, printers, electronic clocks, etc. For example, let's look at any smartphone and a smart TV set. A user may use a smartphone to play selected movies, programmes, music and listen to the news in comfort. There are companies specialising in creation of so called smart homes, where by means of a

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phone or a special controller/center a user can make roller binds go down, turn the light on and off, control the heating level at home by means of a network of sensors and the above mentioned controller. Another example of a certain form of IoT in every-day life is a car equipped with a hands-free set, which couples a phone with a vehicle via bluetooth and enables comfortable search through lists of contacts by a driver and allowed form of phone conversation when driving a car. However, the majority of such systems is limited to communication of two, three devices (save for possible networks of sensors that are treated as subassemblies or elements of a larger project or a device).

But what is IOT [1, 2], that makes it possible for us to benefit from the above mentioned examples. On the basis of the article written by Prajsnar [3] and published on website www.forbes.pl, the Internet of things is most of all an idea enabling transfer of data by means of a transmission mode selected by a designer. Of course, the most popular one is wireless Wi-Fi transmission, that is one of the relatively easiest forms to be implemented. Obviously, there is no rule that the IoT system may be built only on the basis of Wi-Fi communication. There are other forms of data transmission among devices, like the above example of a smartphone connected with a car via bluetooth, or a set with a radio receiver and a radio transmitter. Increasingly more popular term of the Internet of Things seems to be the future in designing of modern devices and according to this idea the device discussed in this article is created.

The presented radio receiver will not be a small module of a radio, clock, alarm clock, but also a mobile version of universal central station of user IOT network that a user himself may develop by adding new functionalities according to his own ideas. The smart home is only one of many examples of possibilities of the project development. This device may be compared at first sight with a smartphone with an access to the Internet, it may be used to benefit from different forms of transmission, such as bluetooth or infrared, thanks to which IOT network may be to a certain extent implemented on the basis of independent devices. But the device presented in the title will be characterised mainly by a significantly lower cost of production as compared to the cheapest versions of smartphones, miniaturised size and larger programmable possibilities for a user.

According to the topic, the main function of our device is to play sound from selected ratio stations and read information from transmitting stations, enabling user's convenient and simple access to information and entertainment – the RDS technology will be implemented in the project [4, 5]. Another essential function of the project is the universal nature of a device in the IOT network as a mobile centre. Other functions, such as a watch, alarm clock, etc. have been described in Subsect. 2.1 containing the project assumptions. We can distinguish 8 typical functions of the discussed RDS technology (which were described in detailed in an article written by Sagan [6]), which define transmitted information:

- Program Services (PS) – name of the station,
- Program Type (PTY) – type of program,
- Radio Text (RT) – any text,
- Clock Time and Date (CT) – current time and date,

- Traffic Program/Traffic Announcement (TP/TA) – information for drivers,
- Program Identification (PI) – station identification code,
- Alternative Frequencies (AF) – list of alternative frequencies,
- Enhanced Other Networks (EON) – information about other programs stations.

On the basis of Krzysztof Sagan's article [7], a definition of radio data system may be presented as a subcarrier with a possibility of its modification by means of digital information, which is added to the traditional broadcasting of very high frequency waves (VHF) with Frequency Modulation (FM); it enables a separation of the data stream by means of duly adjusted receivers and consequently, improved reception and possibility to deliver additional information (which has been listed as RDS functions in the preceding paragraph).

2 Project Description

Universal small radio that can always be at hand. Listening to the news, listening to the music, checking time or setting an alarm. These are the most basic functions, which have occurred at the project designing stage. During the initial works we found out that it may be improved by adding functions, which make it possible for a user to develop the project on its own and add on ideas.

First of all, a user has a screen with OLED display, three micro switches and – what is most important – a coupling enabling transmission of RX and TX signals of the microcontroller, which facilitate use of Wi-Fi and Bluetooth transmission, thanks to which, consequently, a devices may be implemented in any IoT network. A user will dispose with all main functions of our radio receiver, that are described, for instance, below in the assumptions of the project prototype (Subsect. 2.1), while development of the device as the central station in the IOT network is open for the user, meaning that a user has programmable output at his disposal, thanks to which it may upload own programme code responsible for specific needs of ideas.

Obviously, it requires certain programming experience of the user to use fully the potential of a device. The miniaturisation of the device is based mainly on use of a small Lithium-polymer battery, which ensures mobility of a device and use of electronic elements that are adjusted for surface mounting, thanks to which the size of PCB plate will be maximally reduced. Mounting holes enable docing of the project in a selected place, which is one of the additional options.

2.1 Assumptions of the Project Prototype

The following project assumptions have been prepared on the stage of prototyping:

- Receipt of signals sent by UKF FM and RDS stations,
- Displaying information from RDS on OLED display,
- Possibility to play sound by means of an embedded speaker based on mono audio-type amplifier,
- Possibility to play sound via external headset by means of Jack 3.5 mm connection and signal of FM radio receive used,

- Displaying current time, date and temperature on OLED display on the basis of RTC chip,
- A possibility to set an alarm clock, time and date on the basis of RTC chip,
- Output connection for Bluetooth/Wi-Fi communication (a possibility to develop the device to include communication with external devices in IOT network),
- Powering the devices with a small Lithium-Polymer battery, using ATB-LION converter,
- A possibility to charge a device by means of a micro USB able on the basis of ATB-LION charger,
- Intuitive user menu based on OLED display and 3 micro switches and an infrared recipient (receiving signals from a remote device with RC5 coding), which enables setting of time, date, alarms, changing ratio stations,
- Output SPI bus and a connection equipped with RX, TX, signals enables user's modification of the programme through introduction of own changes in respect to handling an OLED display, three micro switches and use of infrared receiver. Moreover, a possibility to introduce own set of AT commands adjusted to communication through Wi-Fi and Bluetooth transmission (receiving and sending information).

2.2 Used Electronic Components in the Prototype

For the prototyping stage, mainly electronic parts adapter for surface mount SMD were used to minimize the prototype as much as possible.

- Microcontroller AVR ATMEGA88-AU, technical documentation by microchip [8],
- RTC system DS3231 – real-time clock system with built-in temperature sensor, maxim integrated technical documentation [9],
- RDA5807 M system – FM radio system with RDS, technical documentation by RDA microelectronics [10],
- TDA7052A system – mono audio amplifier system, NXP technical documentation [11],
- MCP1700T system – voltage stabilizer 3,3 V,
- ATB-LION system – DC-DC converter module (5 V output voltage) and Li-poly battery chargers (maximum 500 mA charging current), atnel technical documentation [12],
- TSOP311236 system – infrared receiver,
- OLED I2C 128x64 display,
- 3.5 mm jack connector,
- Resistors: 0 Ω, 180 Ω, 510 Ω, 10 kΩ, 39 kΩ, 56 kΩ, 100 kΩ, 330 kΩ, 1 M Ω,
- Ceramic capacitors: 22 pF, 1 μF, 10 μF,
- Bipolar capacitors: 22 μF, 100 μF,
- Inductors: 10 nH, 10 μH,
- Transistors: N-MOSFET BSS138 – conversion 5 V do 3,3 V for I2C communication with the RDA5808 M system,
- Quartz: 18.432 MHz,

- 2.54 mm pin connectors – universal RX, TX, VCC, GND line output/SPI transmission,
- Lithium-polymer battery (Li-Poly) 2,8 V–4,2 V.

2.3 Electronic Schemes of the Prototype System

The scheme of the device has been divided into functional parts. Individual of them are presented below:

- Figure 1 presents main block of our device. The electronic components responsible for the power supply are located on the left. On the right side are placed micro-switches, battery outputs and universal output for WIFI or Bluetooth modules, thanks to which the device can operate in IOT networks,

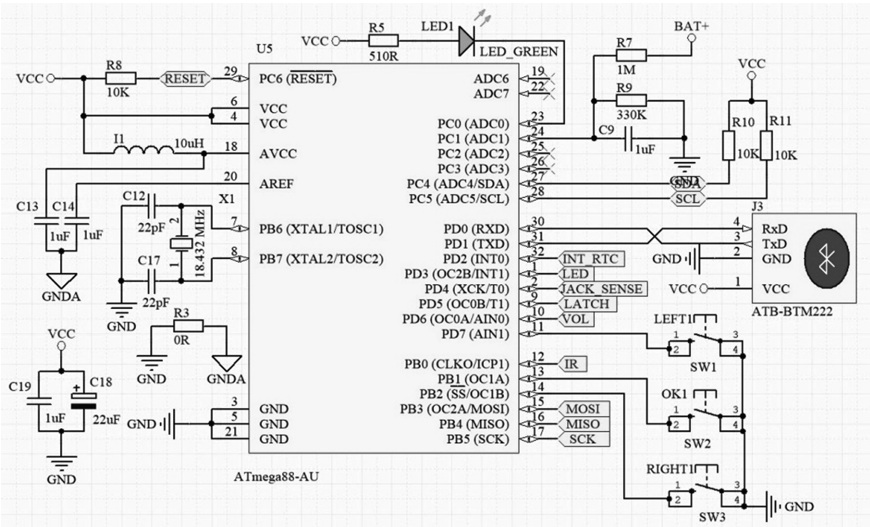


Fig. 1. The scheme for control unit – ATMEGA88

- Figure 2 presents RDS system with simple 5 V – 3.3 V converter based on BSS138 transistors on microcontroller lines – RDS system,

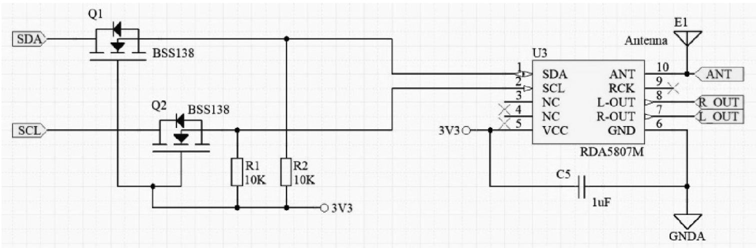


Fig. 2. The scheme for FM radio system with RDS and 5 V to 3.3 V voltage conversion

- Figure 3 presents connection of the system supplying the whole device with 5 V voltage and on the right side a voltage stabilizer 3.3 V,

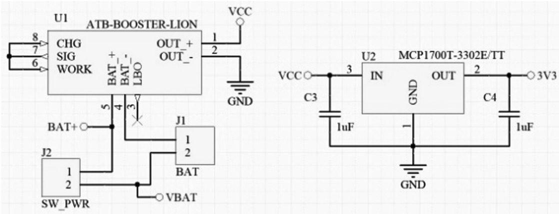


Fig. 3. The scheme for system power supply – 5 V DC/DC converter/charger and 3.3 V voltage stabilizer (for RDA5807M requirements)

- Figure 4 presents connection of the amplifier and headphone system,

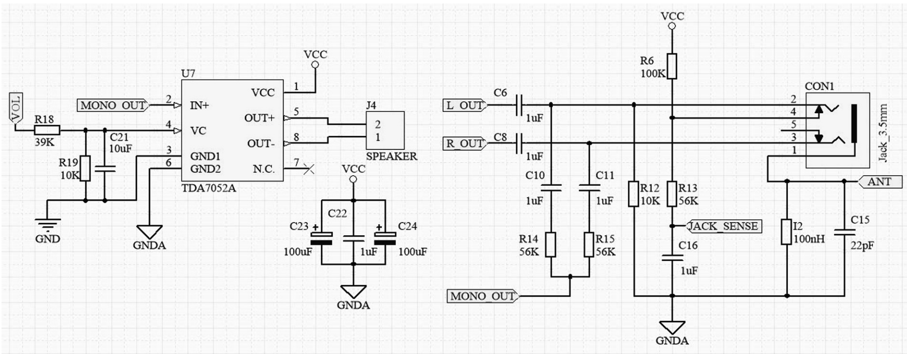


Fig. 4. The scheme for the mono audio amplifier circuit and the headphone input circuit

- Figure 5 presents connection of the RTC system on right side and on the left side SPI program bus output,

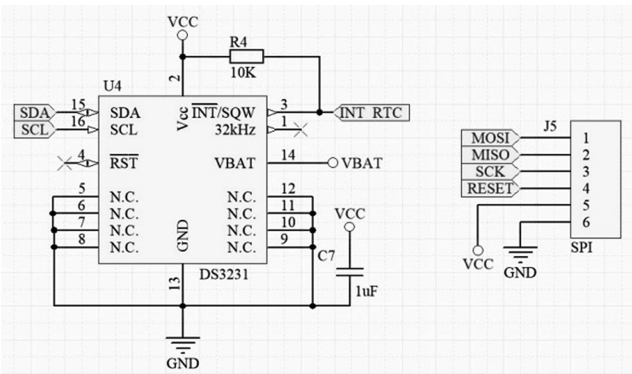


Fig. 5. The scheme for RTC clock chip and SPI program bus output

- Figure 6 presents connection of infrared receiver system and OLED I2C,

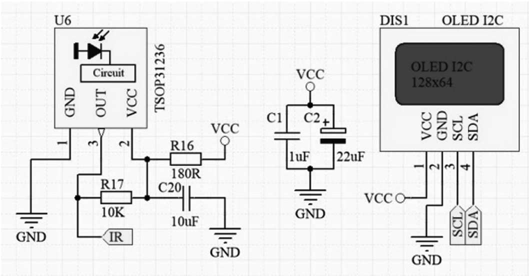


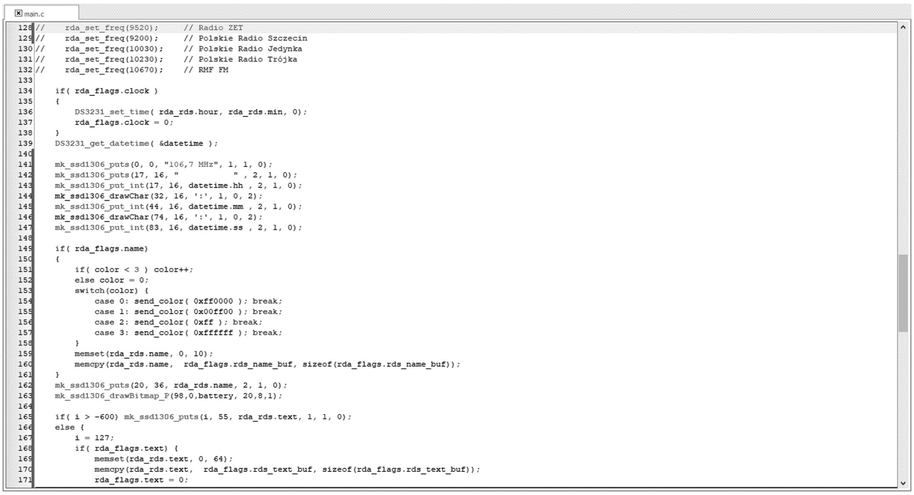
Fig. 6. The scheme for infrared receiver system and OLED I2C display connection

2.4 Operating Principle of the Device

AVR ATMEGA88 microcontroller software has been written in C programming language in Eclipse environment on the basis of books by Kardaś [13–15]. The code is uploaded and the system tested by means of SPI bus and USP-ASP programmer. The programme was written on the basis of the assumptions presented in such-chapter containing the list of functionalities. Communication between microcontroller and the system of RDA5807 M radio receiver is performed through I2C bus. As it can be noticed on scheme 2, ATMEGA88 is powered by 5 V current, while RDA5807 M – by 3.3 V current, scheme 2, 3. To communicate safety between systems powered with different voltage of 5 V and 3.3 V, a simple voltage conversion system should be installed that uses N-MOSFET BSS138-type transistors together with PULL-UP transistors for 3.3 V. The programme operates on the basis of principle of collection of addresses from the memory of radio receiver system, that stores, for instance, current RDS readouts. RDA5807 M system polling takes place in a non-blocking manner, that is on the basis of the programme timer created in the basis of the time of the equipment (TIMERX/COUNTERX). Read information is converted into string-type variables through their uploading to the buffered of characters (CHAR-type variables), and afterwards they are also transmitted by means of I2C bus to the OLED display. There, they are displayed also on the basis of programme timers, while universal menu operates according to the principle of multi-layer display, thanks to which, when creating own programme, a user operates on separate layers of OLED display and therefore when the device operates in the mode of user programme, layers with RDS information, or date, time and alarms may be displayed in intervals set by the user. Moreover, the real time clock (RTC) system is queried every second, which enables downloading of current date, time and temperature. Information is transferred to the display. The user menu may be opened by means of microswitches or infrared remote controllers (RC5 coding mode), and on this level the user may set current time, date, alarm and raise or reduce the volume level (depending on whether headset is connected or not) by means of TDA7052A or RDA5807 M amplifier. Despite entering into and leaving the menu, a user may change frequency – change of the radio station which is received by the device or switch into own programme mode that made operate according to a code written by the user. Communication of the microcontroller with DS3231 system take place according to the same principle as in case of the radio receiver. According to default setting, when it is connected to power, the system emits sounds from a built-in speaker linked with the system of mono audio TDA7052A amplifier. The programme checks on ongoing basis whether headphones are connected to the Jack slot. If it happens, it will entail an event that switches the emission of sound from the built-in speaker linked to an amplifier directly to the RDA5807 M system.

2.5 Fragment of the Source Code

Part of the main loop of the program, responsible for downloading time from the RTC system and displaying of individual RDS contents on the OLED display (Figs. 7 and 8).



main.c

```c
//     rda_set_freq(9520);      // Radio ZET
//     rda_set_freq(9200);      // Polskie Radio Szczecin
//     rda_set_freq(10030);     // Polskie Radio Jedynka
//     rda_set_freq(10230);     // Polskie Radio Trójka
//     rda_set_freq(10670);     // RMF FM

    if( rda_flags.clock )
    {
        DS3231_set_time( rda_rds.hour, rda_rds.min, 0);
        rda_flags.clock = 0;
    }
    DS3231_get_datetime( &datetime );

    mk_ssd1306_puts(0, 0, "106,7 MHz", 1, 1, 0);
    mk_ssd1306_puts(17, 16, "           " , 2, 1, 0);
    mk_ssd1306_put_int(17, 16, datetime.hh , 2, 1, 0);
    mk_ssd1306_drawChar(32, 16, ':', 1, 0, 2);
    mk_ssd1306_put_int(44, 16, datetime.mm , 2, 1, 0);
    mk_ssd1306_drawChar(74, 16, ':', 1, 0, 2);
    mk_ssd1306_put_int(83, 16, datetime.ss , 2, 1, 0);

    if( rda_flags.name)
    {
        if( color < 3 ) color++;
        else color = 0;
        switch(color) {
            case 0: send_color( 0xff0000 ); break;
            case 1: send_color( 0x00ff00 ); break;
            case 2: send_color( 0xff ); break;
            case 3: send_color( 0xffffff ); break;
        }
        memset(rda_rds.name, 0, 10);
        memcpy(rda_rds.name,  rda_flags.rds_name_buf, sizeof(rda_flags.rds_name_buf));
    }
    mk_ssd1306_puts(20, 36, rda_rds.name, 2, 1, 0);
    mk_ssd1306_drawBitmap_P(98,0,battery, 20,8,1);

    if( i > -600) mk_ssd1306_puts(i, 55, rda_rds.text, 1, 1, 0);
    else {
        i = 127;
        if( rda_flags.text) {
            memset(rda_rds.text, 0, 64);
            memcpy(rda_rds.text,  rda_flags.rds_text_buf, sizeof(rda_flags.rds_text_buf));
            rda_flags.text = 0;
```

Fig. 7. A fragment for the source in C – main.c

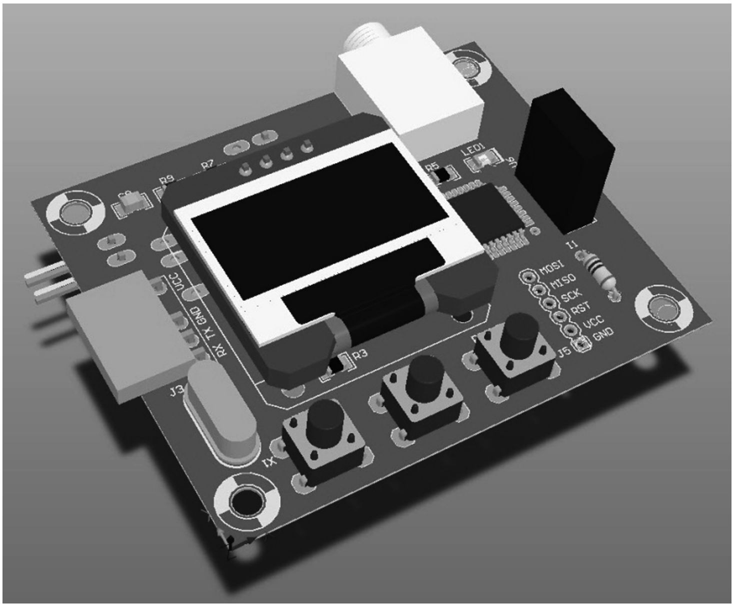


Fig. 8. Prototype – 3D model

2.6 Device Preview in 3D View

The prototype of our device in the design stage.

3 Summary

The device presented in the title may be in a way competitive to smartphones, stationary radio receivers, companies specialising in creation of IOT-related projects, and to launch and educational sets. The table below (Table 1) presents a comparison of our device to the average smartphone.

As compared to the companies on the market that deal with the above mentioned topics of the smart home and IOT in this respect (Table 1 - Usability in IOT networks), a new undertaking would be likely to occur due to smaller costs of production (Table 1 – Production cost) of a smart radio receiver and a base with tutorials that could present certain paths to be used in order to develop the project on one's own, while learning new programming languages.

The programme was written in a non-blocking form (using device and programme timers). Therefore, the user is free to introduce changes to the programme on its own (Table 1 – Possibility to modify the source code by user). It is an advantage of selected functional systems, to be used for the project, such as: DS3231, OLED display and RDA5807 M that they communicate with the microcontroller by means of I2C bus. Such solution makes it possible to save memory in respect to the volume of the code (handling one communication protocol for three systems).

This fact significantly affects the entire project, because a user, who wants to benefit from the entire potential of the device, can also use as much of free memory as possible.

Table 1. Comparison of the title device to the average smartphone

Functionalities	Our device	Average smartphone
Production cost	Low (~30–40 USD)	High (30–1000 USD)
Dimensions	Small (5 × 5 × 2 cm)	Medium (10 × 6 × 0.7 cm)
Possibility to modify the source code by user	Yes	Impossible
RDS radio function	Available immediately	User need to download and install dedicated application
Difficulty level of modifying code by user	Easy (knowledge of the basics of the C language)	Medium (knowledge of java, css, html, application development)
Usability in IOT networks	High (user is able to modify source code for his needs)	Medium (user can only create application)

Of course, every technological solution mentioned above has its strengths and weaknesses, each of them offers certain possibilities to the user, and limits certain solutions, devices have different dimensions and are based on different technology. But the device presented in the title relies on innovation, which means the user has a possibility to develop the project on his own.

Authors think about the further development of the projects as if the device enters into the market, an independent educational platform may be added to the project in form of an Internet website with descriptions, articles, instructions and video tutorials about creation and programming of devices for instance for the smart home that is so popular now. In such case, the entire project would gain an additional feature like educational for beginners in learning programming, which can highly influence on popularity of our project.

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